

ADVANCED COST ACCOUNTING I

ACC 312

Cost: Cost is a measurement, in monetary terms, of the amount of resources used for the purpose of production of goods or rendering services. Cost in simple, words, means the total of all expenses. Cost is also defined as *the amount of expenditure (actual or notional) incurred on or attributable to a given thing or to ascertain the cost of a given thing.* Thus it is that which is given or in sacrificed to obtain something.

The cost of an article consists of actual outgoings or ascertained charges incurred in its production and sale. Cost is a generic term and it is always advisable to qualify the word cost to show exactly what it meant, e.g., prime cost, factory cost, etc. Cost is also different from value as cost is measured in terms of money whereas value in terms of usefulness or utility of an article.

Cost Accounting : Cost Accounting may be defined as “Accounting for costs classification and analysis of expenditure as will enable the total cost of any particular unit of production to be ascertained with reasonable degree of accuracy and at the same time to disclose exactly how such total cost is constituted”. Thus Cost Accounting is classifying, recording an appropriate allocation of expenditure for the determination of the costs of products or services, and for the presentation of suitably arranged data for the purpose of control or guidance of management.

Thus Cost Accounting is a quantitative method that collects, classifies, summarises and interprets information for product costing, operation planning and control and decision making. Therefore, cost accounting involves

- The application of costing principles, methods and techniques
- The ascertainment of cost
- Analysis of servings or excesses

Objectives of Cost Accounting

The following are the main objectives of Cost Accounting :-

- (a) To ascertain the Costs under different situations using different techniques and systems of costing
- (b) To determine the selling prices under different circumstances
- (c) To determine and control efficiency by setting standards for Materials, Labour and Overheads
- (d) To determine the value of closing inventory for preparing financial statements of the concern
- (e) To provide a basis for operating policies which may be determination of Cost Volume market, whether to close or operate at a loss, whether to manufacture or buy from improved method of production....etc.
- (f) To achieve real and permanent reduction in the unit cost of goods manufactured or services rendered without impairing their suitability for the use intended or diminution in the quality of the product.

Scope of Cost Accounting

The scope of Cost Accounting is very wide and includes the following:-

- (a) **Cost Ascertainment:** The main objective of Cost Accounting is to find out the Cost of product / services rendered with reasonable degree of accuracy.
- (b) **Cost Accounting:** It is the process of Accounting for Cost which begins with recording of expenditure and ends with preparation of statistical data.
- (c) **Analysis of Cost:** It is the process of locating the factors responsible for difference in actual cost from the budgeted costs and fixing up of responsibility for differences in cost. It provides better cost management and helps in taking strategic decisions.
- (d) **Cost Control:** It is the process of regulating the action so as to keep the element of cost within the set parameters.
- (e) **Cost Reports:** This is the ultimate function of Cost Accounting. These reports are primarily prepared for use by the management at different levels. Cost reports helps in planning and control, performance appraisal and managerial decision making.
- (f) **Cost Audit:** Cost Audit is the verification of correctness of Cost Accounts and check on the adherence to the Cost Accounting plan. Its purpose is not only to ensure the arithmetic accuracy of cost records but also to see the principles and rules have been applied correctly.

Advantages of cost accounting

- (i) A cost system reveals unprofitable activities, losses or inefficiencies occurring in any form such as
- (a) Wastage of man power, idle time and lost time.
- (b) Wastage of material in the form of spoilage, excessive scrap etc., and
- (c) Wastage of resources, e.g. inadequate utilization of plant, machinery and other facilities.
- (ii) Cost Accounting locates the exact causes for decrease or increase in the profit or loss of the business. It identifies the unprofitable products or product lines so that these may be eliminated or alternative measures may be taken.
- (iii) Cost Accounts furnish suitable data and information to the management to serve as guides in making decisions involving financial considerations.
- (iv) Cost Accounting is useful for price fixation purposes. Although sale price is generally related more to economic conditions prevailing in the market than to cost, the latter serves as a guide to test the adequacy of selling prices.
- (v) With the application of Standard Costing and Budgetary Control methods, the optimum level of efficiency is set.
- (vi) Cost comparison helps in cost control. Comparison may be period to period, of the figures in respect of the same unit or factory or of several units in an industry by employing Uniform Costs and Inter-Firm Comparison methods. Comparison may be made in respect of cost of jobs, process or cost centres.
- (vii) A cost system provides ready figures for use by the Government, wage tribunals and boards, and labour and trade unions.

Disadvantages of cost accounting

Like any other system of accounting, Cost Accounting is not an exact science but an art which has developed through theories and accounting practices based on reasoning and common sense. Many of the theories cannot be proved nor can they be disproved. They grow up in course of time to become conventions and accepted principles of Cost Accounting. These principles are by no means static, they are changing from day to day and what is correct today may not hold true in the circumstances tomorrow.

Steps in installing a Cost Accounting System

1. Establish the objectives and scope of the proposed system through discussions and agreement with management
2. (a) Analyse the current systems:
 - are there cost centers?
 - Is there a coding system?
 - What accounting records are maintained by the existing system etc.(b) study the current staff:
 - staff composition by qualification, experience and competence
 - production system and methods:
 - < is it a jobbing or engineering business?
 - < is it contract business?
 - < Is it service provision ? etc
 - Stores keeping systems
 - Wages systems etc
3. Start to think about the type and nature of the cost accounting system required
 - Is it a job costing system or batch costing system
 - Is it a contract costing system
 - Is it a process costing system
 - Is it a service costing system
 - What approach or technique of costing is required? Full absorption costing, marginal or standard costing?
 - What method will be used to price stores issues ?
 - What method of remuneration are desirable?
 - What will be the basis of overhead absorption? Etc
4. Design the system in collaboration with management and the staff who will operate it.
5. Prepare an operating manual for all personnel showing procedures, objectives and codes.
6. Arrange for training and instruction for all staff who will be involved in implementing the system.
7. Implement the system with full publicity.
8. Monitor the system closely and periodically report to management.

Problems inherent in the installation of a cost accounting system

1. Size of the organization
 2. Type of product/services
 3. The production process
 4. The methods of manufacture
 5. Availability of staff
- Other specific problems include
- Definition of responsibility
 - Definition/designation of cost centres
 - Compilation of comprehensive cost system

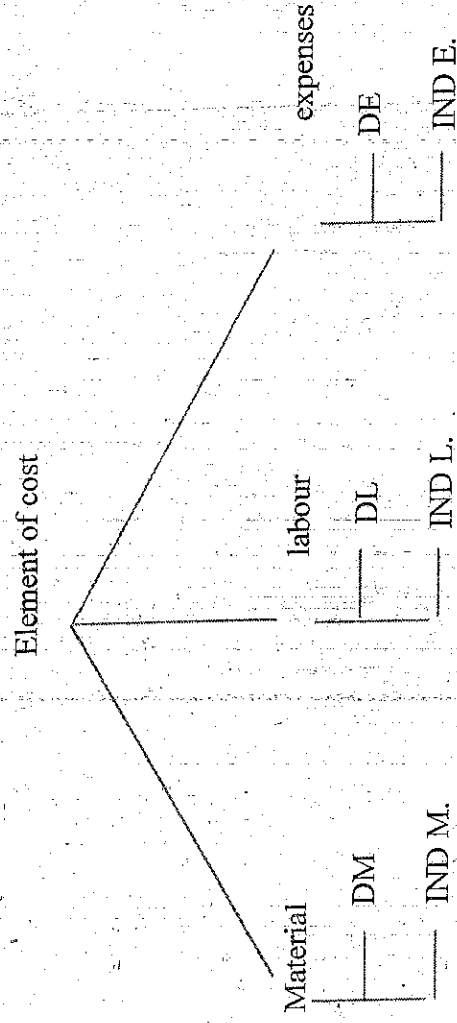
- Availability of staff:
- < labour hours worked
- < machine utilization time
- < scrap
- < rectification cost etc
- The difficulty of accurately classifying cost

ELEMENTS OF COST

There are basically three elements of cost, namely materials, labour and expenses which results to the following cost

9. Material cost
10. Labour cost
11. Expenses.

These are further sub divided into Direct Material and indirect materials, Direct Labour and Indirect Labour Direct, and direct Expenses and Indirect Expenses



Direct Material + Direct Labour + Direct Expenses = Prime Cost
 Indirect Material + Indirect Labour + Indirect Expenses = Overheads

Direct Material Cost

Direct material cost can be defined as *'The Cost of material which can be attributed to a cost object in an economically feasible way'*. Direct materials are those materials which can be identified in the product and can be conveniently measured and directly charged to the product. Thus, these materials directly enter the product and form a part of the finished product. For example, timber in furniture making, cloth in dress making, bricks in building a house. The following are normally classified as direct materials :-

- (i) All raw materials, like animal skin in the manufacture of bags and fruits in canning industry
- (ii) Materials specifically purchased for a specific job, process or order, like glue for book binding, starch powder for dressing yards
- (iii) Parts or components purchased or produced, like batteries for transistor-radios
- (iv) Primary packing materials like cartons, wrappings, card-board boxes, etc.

Indirect Material Cost

Indirect materials are those materials which do not normally form a part of the finished product. It has been defined as "materials which cannot be allocated but which can be apportioned to or absorbed by cost centres or cost units". These are:

- (i) Stores used in maintenance of machinery, buildings, etc., like lubricants, cotton waste, bricks and cements
- (ii) Stores used by the service departments, i.e., non-productive departments like Power House, Boiler House and Canteen, etc., and
- (iii) Materials which due to their cost being small, are not considered worthwhile to be treated as direct materials.

Direct Labour / Employee Cost

The cost of employees can be attributed to a cost object in an economically feasible way. In simple words, it is that labour which can be conveniently identified or attributed wholly to a particular job, product or process or expended in converting raw materials into finished goods. Wages of such labour are known as direct wages. Thus it includes payment made to the following groups of labour:

- (i) Labour engaged on the actual production of the product or in carrying out of an operation or process
- (ii) Labour engaged in adding the manufacture by way of supervision, maintenance, tool setting, transportation of material etc.
- (iii) Inspectors, analysts etc., specially required for such production

Indirect Labour/ Employee Cost

The labour / employee cost cannot be directly attributed to a particular cost object. The wages of that labour which cannot be allocated but which can be apportioned to or absorbed by cost centres or cost units is known as Indirect Labour. Example of such labour are: charge-hands and supervisors; maintenance workers; men employed in service departments, material handling and internal transport; apprentices, trainees and instructors; clerical staff and labour employed in time office and security office.

Direct or Chargeable Expenses

Direct expenses are expenses relating to manufacture of a product or rendering a service which can be identified or linked with the cost object other than direct material cost and direct employee cost. Direct expenses include all expenditure other than direct material or direct labour that is specifically incurred for a particular product or process. Such expenses are charged directly to the particular cost account concerned as part of the prime cost. Examples of direct expenses are: (i) GST; (ii) Royalty; (iii) Architect or Supervisor's fees; (iv) Cost of rectifying defective work; (v) Travelling expenses to the city; (vi) Experimental expenses of pilot projects; (vii) Expenses of designing or drawings of patterns or models; (viii) Repairs and maintenance of plant obtained on hire; and (ix) Hire of special equipment obtained for a contract.

Overhead

Overheads comprise of indirect materials, indirect employee cost and indirect expenses which are not directly identifiable or allocable to a cost object. Overheads may be defined as the aggregate of the cost of indirect material, indirect labour and such other expenses including services as cannot conveniently be charged directly to specific cost units. Thus overheads are all expenses other than direct expenses. In general terms, overheads comprise all expenses

incurred for or in connection with, the general organization of the whole or part of the undertaking, i.e., the cost of operating supplies and services used by the undertaking and includes the maintenance of capital assets.

Prime Cost

The aggregate of Direct Material, Direct Labour and Direct Expenses is called Prime Cost. Generally it constitutes 50% to 80% of the total cost of the product, as such, as it is primary to the cost of the product and called Prime Cost.

Cost Object : Cost object is the technical name for a product or a service, a project, a department or any activity to which a cost relates. The Cost object could be defined broadly or narrowly. At a broader level a cost object may be named as a *Cost Centre*, where as at a lowermost level it may be called as a *Cost Unit*.

Cost Driver : CIMA terminology defines a Cost Driver as "the factor influencing the level of cost, often used in the context of ABC to denote the factor which links activity resource consumption to produce outputs, for example, the number of purchase orders would be a Cost Driver for procurement cost". In other words, we can say that, the Cost Driver is an activity which is responsible for cost incurrence. An activity may be an event, unit of work etc. Examples of the Cost Drivers are (i) number of machine set-ups, (ii) number of purchase orders, (iii) hours spent on product inspection, (iv) number of tests performed etc.

Cost Centre : CIMA defines a cost centre as "a location, a person, or an item of equipment (or a group of them) in or connected with an undertaking, in relation to which costs ascertained and used for the purpose of cost control". The determination of suitable cost centres as well as analysis of cost under cost centres is very helpful for periodical comparison and control of cost. The selection of suitable cost centres or cost units for which costs are to be ascertained in an undertaking depends upon a number of factors such as organization of a factory, condition of incidence of cost, availability of information, requirements of costing and management policy regarding selecting a method from various choices. Cost centre may be production cost centres operating cost centres or process cost centres depending upon the situation and classification.

Economic Order Quantity: (EOQ)

The total costs of a material usually consist of Buying Cost + Total Ordering Cost + Total Carrying Cost.

Economic Order Quantity is 'The size of the order for which both ordering and carrying cost are minimum'.

Ordering Cost: The costs which are associated with the ordering of material. It includes cost of staff posted for ordering of goods, expenses incurred on transportation, inspection expenses of incoming material....etc

Carrying Cost: The costs for holding the inventories. It includes the cost of capital invested in inventories. Cost of storage, Insurance.....etc

The assumptions underlying the Economic Ordering Quantity (EOQ): The calculation of economic order of material to be purchased is subject to the following assumptions:-

- (a) Ordering cost per order and carrying cost per unit per annum are known and they are fixed.
- (b) Anticipated usage of material in units is known.
- (c) Cost per unit of the material is constant and is known as well.

(d) The quantity of material ordered is received immediately i.e. lead time is Zero. The famous mathematician 'WILSON' derived the formula used for determining the size of order for each purchases at minimum ordering and carrying costs, which is as below :-

Economic Ordering Quantity =

$$\sqrt{\frac{2AO}{C}}$$

Where,

A = Annual demand /Consumption

O = Ordering Cost per order

C = Carrying Cost per unit per annum

Illustration 1

Calculate the Economic Order Quantity from the following information. Also state the number of orders to be placed in a year.

Consumption of materials per annum : 10,000 kg

Order placing cost per order : # 50

Cost per kg. of raw materials : # 2

Storage costs : 8% on average inventory

Solution:

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}}$$

C = Inventory carrying cost per unit per annum $2 \times 8\% = \text{' } 0.16$

EOQ = $\frac{2 \times 10,000 \text{ (units)} \times 50}{0.16}$

EOQ = 2,500 kg.

No. of orders to be placed in a year = $\frac{\text{Total consumption of materials per annum}}{EOQ}$

$$= \frac{10,000 \text{ kg}}{2,500 \text{ kg}}$$

= 4 Orders per year

Illustration 2

The average annual consumption of a material is 18,250 units at a price of #36.50 per unit. The storage cost is 20% on an average inventory and the cost of placing an order is # 50. How much quantity is to be purchased at a time?

Solution:

EOQ = $\frac{2 \times 18,250 \text{ (units)} \times 50}{C}$

C = 20% of 36.50 = 7.3

$$= \frac{1,825,000}{7.3}$$

= 250 Units

Material Storage & Control

Once the material is received, it is the responsibility of the stores-in-charge, to ensure that material movements in and out of stores are done only against the authorized documents. Stores-in-charge is responsible for proper utilization of storage space & exercise better control over the material in the stores to ensure that the material is well protected against all losses such as theft, pilferage, fire, misappropriation ...etc.

Duties of store keeper

The duties of store-keeper are as follows :-

- (a) To exercise general control over all activities in stores department
- (b) To ensure safe storage of the materials
- (c) To maintain proper records
- (d) To initiate purchase requisitions for the replacement of stock of all regular materials, whenever the stock level of any item in the store reaches the Minimum Level
- (e) To initiate the action for stoppage of further purchasing when the stock level approaches the Maximum Level
- (f) To issue materials only in required quantities against authorized requisition documents
- (g) To check and receive purchased materials forwarded by the receiving department and to arrange for storage in appropriate places

Classification and Codification of Material

In case of large organizations the number and types of materials used is considerable and unless each item is distinguished and stored separately it would be impossible to find them out when they are required for production or any other operation.

There are two steps in the classification and codification of materials - determination of the number of items, their nature, other characteristics and classification of the items of comparable nature or type into suitable groups or classes.

Various classes of coding are in practice and the common types are : **Alphabetical Scheme** and **Numeric Scheme**. Others include decimal scheme, block scheme and combination scheme.

Advantages of Classification & Codification of materials

- (a) The procedure assists in the easy identification and location of the materials because of their classification.
 - (b) It minimises the recording of the nature/ type of the materials with detailed description on every document relating to the transaction of materials.
 - (c) Codification is a must in the case of mechanisation of the stores accounting.
 - (d) The method is simple to operate and definitely saves time and money in respect of both physical location/identification of materials as well as recording of the materials.
- After the material classification and codification is done for all the materials, for each material code we have to fix the Minimum Level, Maximum Level, Re-order Level and Re-order Quantity. It is the storekeeper's responsibility to ensure inventory of any material is maintained between the Minimum Level and Maximum Level.

MAXIMUM LEVEL: The Maximum Level indicates the maximum quantity of an item of material that can be held in stock at any time. The stock in hand is regulated in such a manner that normally it does not exceed this level.

While fixing the level, the following factors are to be taken into consideration:

- (a) Maximum requirement of the store for production purpose, at any point of time
- (b) Rate of consumption and lead time

- (c) Nature and properties of the Store: For instance, the maximum level is necessarily kept low for materials that are liable to quick deterioration or obsolescence during storage.
 - (d) Cost of storage and insurance
 - (e) Economy in prices: For seasonal supplies purchased in bulk during the season, the maximum level is generally high.
 - (f) Financial considerations: Availability of funds and the price,
 - (g) Rules framed by the government for import or procurement.
 - (i) The maximum level is also dependent on the economic ordering quantity.
- Maximum Level = Re-Order Level + Re-Order Qty – (Minimum Rate of Consumption × Minimum Re-Order Period)**

MINIMUM LEVEL : The Minimum Level indicates the lowest quantitative balance of an item of material which must be maintained at all times so that there is no stoppage of production due to the material being not available. In fixing the minimum level, the following factors are to be considered :-

- (a) Nature of the item: For special material purchased against customer's specific orders, no minimum level is necessary. This applies to other levels also.
 - (b) The minimum time (normal re-order period) required replenishing supply: This is known as the Lead Time and are defined as the anticipated time lag between the dates of issuing orders and the receipt of materials. Longer the lead time, lower is minimum level, the re-order point remaining constant.
 - (c) Rate of consumption (normal, minimum or maximum) of the material.
- Minimum Level = Re-Order level – (Normal Rate of Consumption × Normal Re-Order Period)**
- Re-Order Level**

REORDERING LEVEL : When the stock in hand reach the ordering or re-ordering level, store keeper has to initiate the action to replenish the material. This level is fixed somewhere between the maximum and minimum levels in such a manner that the difference of quantity of the material between the Re-ordering Level and Minimum Level will be sufficient to meet the requirements of production up to the time the fresh supply of material is received.

The basic factors which are taken into consideration in fixing a Re-ordering Level for a store item include minimum quantity of item to be kept, rate of consumption and lead time which are applied for computing of this level.

Re-Ordering level = Minimum Level + Consumption during lead time

= Minimum Level + (Normal Rate of Consumption × Normal Re-order Period)

Another formula for computing the Re-Order level is as below:

Re-Order level = Maximum Rate of Consumption × Maximum Re-Order period (lead time)

DANGER LEVEL : It is the level at which normal issue of raw materials are stopped and only emergency issues are only made. This is a level fixed usually below the Minimum Level. When the stock reaches this level very urgent action for purchases is indicated. This presupposed that the minimum level contains a cushion to cover such contingencies.

Danger Level = Normal Rate of Consumption × Maximum Reorder Period for emergency purchases.

Illustration 3

The components A and B are used as follows:

Normal usage ... 300 units per week each

Maximum usage 450 units per week each
 Minimum usage 150 units per week each
 Reorder Quantity A 2,400 units; B 3,600 units.
 Reorder period A 4 to 6 weeks, B 2 to 4 weeks.
 Calculate for each component:

(a) Re-order Level (b) Minimum Level (c) Maximum Level (d) Average Stock Level.

Solution:

Particulars

	A	B
a)	Reorder Level [Max. Consumption $\times$ (450 $\times$ 6) = Max. Re-order Period] 2700 units	(450 $\times$ 4) = 1800 units
b)	Minimum Level [ROL - (Normal Consumption $\times$ Re-order period)] [2700 - (300 $\times$ 5)] = 1200 units	[1800 - (300 $\times$ 3)] = 1200 units
c)	Maximum Level [ROL + ROQ - (Min. Consumption $\times$ Min. Re-order Period)] [2700 + 2400] - (150 $\times$ 2) = 5100 units	[1800 + 3600 - (150 $\times$ 2)] = 5100 units
d)	Average Stock Level [Min. Level + Max. Level] / 2 OR [Min. Level + $\frac{1}{2}$ Re-order Quantity] 2850 units [4500 + 1200]/2 (or) 2400 units 1200 + $\frac{1}{2}$ (2400)	3000 units [5100 + 900]/2 (or) 2700 units 900 + $\frac{1}{2}$ (3600)

INTEGRATION OF COST AND FINANCIAL ACCOUNTS

- Book keeping
- Ledger accounts with interlocking
- Integrated ledger accounts
- Reconcile accounts and cost accounting profits.

Introduction

Book keeping is the process of identifying, recording, and summarizing transactions in the appropriate books of accounts. The balances in the accounts, to be obtained at the end of an accounting period, are to be extracted to the trial balance, which is just list of balances of both debit and credit. The system of book keeping is not restricted to Financial Accounting; it is applicable to Cost Accounting too. Both financial and cost accounting data are to be properly kept and processed to develop and produce the required information for decision making in the future.

The size of a business normally determines the complexity of the cost records. Small businesses keep simple records not linked to the financial accounts, while larger businesses maintain a full set of cost accounts which provides the details behind the totals shown in the financial accounts. In some firms, the cost accounts will be kept independent of the financial accounts and periodic reconciliation will be necessary. In other firms, the cost accounts form part of the formal accounting system and no reconciliation is necessary. The latter method is more satisfactory because it avoids duplication of effort and reconciliation problems.

INTERLOCKING ACCOUNTING SYSTEM AND COST LEDGER

This accounting system uses separate cost accounts which periodically are reconciled with the financial accounts.

The cost accountant must use a great deal of the information which is also available to the financial accounts, e.g. purchase day book, expenses daybook, payroll, main journal, sales daybook, etc.

The structure of the cost accounts will be formed according to the complexity of the business and will often include accounts for: cost elements, cost centres (production, marketing and general administration), job cost (production, repairs, capital) and stock (i.e. raw materials, components or spare parts, finished goods). The interlocking of the two systems is carried out by the use of control accounts in each set of accounts, i.e.

a. cost ledger control account in the financial ledger b. financial ledger control account in the cost ledger.

The Cost Ledger: The cost ledger is the principal ledger in the cost book-keeping system when the system is separated from the financial accounts. Generally, the cost ledger will contain:

(a) Control accounts for stores, work-in-progress, finished goods and overhead. Each control account will be supported by a subsidiary ledger to provide the necessary details for accounting control and preparation of information.

(b) Accounts for wages, sales, cost of sales, together with any other account needed in the cost book-keeping system, e.g. Notional charges, Suspense accounts.

(a) A Cost Ledger Contra Account. This account is needed to maintain the double-entry principle within the cost ledger and contains postings which affect accounts outside the costing system, e.g. Debtors, Creditors or Cash Accounts of Real and Personal accounting transactions. This ledger is also referred to as :

i. General Ledger Adjustment Account; and

ii. Cost Ledger Control Account

The focal point of the cost accounting system is the work-in-progress ledger. In a specific order business, the ledger will contain a separate account for each job and will be debited with the cost of material used, wages allocated and production overhead absorbed. The cost of completed jobs is transferred to Finished Goods Ledger. When the job is delivered to the customer, cost of sales will be debited with the Job Cost and Sales credited with the selling price. Thus, a (costing) profit and loss statement can be prepared and analysed by job. The balances on the control accounts represent the cost value of stores, work-in-progress and finished goods.

Summary of Relevant Cost Accounting entries under interlocking system up to WIP Control Account.

1. For Raw Material Incurred
Debit Raw Material Control A/c Or Stores Ledger Control Account
Credit Cost Ledger Control Account/General Ledger Adjustment Account/
Cost Ledger Control Account

2. For Materials returned to Suppliers
Debit Cost Ledger Control Account or General Ledger Adjustment Account
Credit Raw Materials Control Account or Stores Ledger Control Account

3. For Material Issued
Debit Work-in-Progress Control Account, WIP (Direct)
Debit Production Overhead Control Account, POCA (Indirect
Material issued to factory)
Credit Raw Materials Control Account

4. For Wages Incurred
Debit Wages Control Account, WCA
Credit Cost Ledger Control Account

5. For Wages Allocated
Debit Work-in-Progress Control A/C (Direct)
Debit Production Overhead Control A/c (Indirect to factory)
Credit Cost Ledger Control Account

6. For Production Overhead Absorbed
Debit WIP Control Account
Credit Production Overhead Control Account

Note: Adjustment entry is made for under or over absorption of overhead

7. For Output Achieved:
Debit Finished Goods Accounts
Credit Work-in-Progress Control Account

RECONCILIATION OF COSTING AND FINANCIAL ACCOUNTING PROFITS

The profit disclosed by the Costing Profit and Loss Account may differ from that shown in the financial Profit and Loss Account where interlocking system is adopted and it is necessary to reconcile them. The difference may be due to the following reasons:

Appropriations of Profit not Dealt With in the Cost Accounts

- (a) Corporate tax
- (b) Transfers to general reserve or any other fund of accumulated profits, e.g. dividend equalization reserve.
- (c) Charitable donations, where no direct benefit is derived therefrom by the company's employees.
- (d) Dividends paid on the share capital of the company
- (e) Additional provisions for depreciation of buildings, plants, etc., and for bad debts.
- (f) Amounts written off on goodwill, preliminary expenses, underwriting commission, debenture discounts, expenses of capital issues, etc.
- (g) Appropriations to sinking funds for the purpose of providing for the repayment of loans or debentures.

Financial Matters that are Outside the Scope of Cost Accounts

- (a) Interest received on bank deposits and investments.
- (b) Rents receivable. If, however, rents are received from a portion of the business premises which has been sublet, the profit on rental only should be excluded from the cost records. The proportion which represents recovery of rent paid will be deducted from total rent to ascertain the net cost of the premises to be included in the cost records.
- (c) Profits or losses on sales of investments, buildings and other fixed assets; capital expenditure charged specifically to revenue.
- (d) Interest on bank loans, mortgages, debentures and other borrowed money, if interest on capital is ignored in the cost records.
- (e) Damages payable according to law.
- (f) Penalties payable for late completion of contracts.
- (g) Losses due to scrapping of machinery before the date estimated when depreciation and obsolescence were provided for.

ILLUSTRATION 1

NUASA Limited has separate financial and cost accounting systems. Extracts from both sets of accounts for the year ended 31 December 2011 were as follows:

(a) Financial Accounts

	Opening Stock #	Closing Stock #
Stock valuations		
Raw materials	34,000	36,000
Work-in-progress	7,000	5,000
Finished goods	16,000	20,500
Debtenture interest		5,000
Interest received		750
Discounts allowed		2,000
Discounts received		1,000
Net Profit (before taxation)		17,500

(b) Cost Accounts

Opening Stock Closing Stock	#	#
Stock valuations		
Raw materials	33,000	34,500
	13	

Work-in-progress	8,000	7,000
Finished goods	17,000	20,500

Required:

Reconcile the profit as reported in the financial accounts with the cost accounts, and calculate the cost accounting profit for the *period*.

INTEGRATED ACCOUNTING SYSTEM

In this accounting system, the concept of separate Profit and Loss Accounts for financial and cost purposes is discarded in favour of a unified set of accounts which serves both financial and management purposes. Such a system of accounting is referred to as "integrated" or "integral" system. The terminology (CIMA) defines integrated accounts as "set of accounting records that integrates both financial and cost accounts using a common input and data for all accounting purposes." The system is designed on the following principles:

- (a) The degree of integration must be determined. Some undertakings find it satisfactory merely to integrate up to the stage of prime cost, leaving overhead allotment to be dealt with in memorandum form only.
- (b) Dependent upon the degree of integration decided upon, the normal classification of expenditure by nature is discarded
- (c) Full details of items posted to the control accounts are supplied to the cost department at conventional intervals, accompanied by prelists or registers of amounts to ensure completeness.
- (d) The information is then dealt with by the cost department in accordance with the system of costing in force and with the following objectives in mind:
 - (i) To provide the necessary costing data.
 - (ii) To form the basis for journal entries so that the control accounts can be cleared to suitable revenue accounts, culminating in a profit and loss account.

Entries for Integrated System

1. For Materials Purchased
Debit Raw Materials/Stores Ledger Control A/c
Credit Bank/Creditors for Materials.
2. For Materials returned to Suppliers
Debit Creditors for Materials Indirect Factory)
Credit Raw Materials Control Account
3. For Materials Issued
Credit Raw Materials Control Account
Debit Work-in-Progress Control Account (Direct)
Debit Production Overhead Control Account (
4. For Wages Incurred:
Debit Wages Control Account, WCA
Credit Wages Payable Account
5. For Wages Allocated
Debit Work-In-Progress Control Account
Credit Production Overhead Control Account
Credit Wages Control Account
6. For Production Overhead Incurred
Debit Production Overhead Control Account, POCA
Credit Bank/Creditors for Overheads

7. For Absorption of Overhead
Debit WIP Control Account
Credit Production Overhead Control Account
Note: Adjust for under/over absorption

8. For Output Achieved
a. Finished Goods Account
b. Credit Work-In-Progress Account

ILLUSTRATION 2

Gwammaja Limited manufactures a range of products which are sold through a network of wholesalers and dealers. A set of integrated accounts is kept, and for the year 2011 the following information is relevant:

(a) Production overhead is absorbed into the cost of products on the basis of a budgeted rate of 80% of direct labour cost.

(b) Finished stocks are valued at factory cost.

	31 march, 2011	30 April, 2011
	#	#
Raw materials stock	8600	7580
WIP	2800	2375
Finished goods stock	5250	6045
Debtors for goods sold	4600	5570
Creditors for raw materials	3800	4710
Fixed asset @ net book value	3000	3000

(d) Bank transactions for the month of April 2011 were as follows:

Bank balance at 31st March	# 750
Receipts from debtors	# 13,780
Payments made:	
Direct labour	# 3,200
Creditors for raw materials	# 4,480
Production overhead	# 2,700
Administration overhead	# 350
Selling & Distribution overhead	# 1,150

Required:

- (a) Use the information above to write up the following control accounts: (i) Raw materials stock
(ii) Work-in-Progress
(iii) Finished Goods
(iv) Production Overhead

COSTING METHODS

This is the method that is used to determine the amount or value of the economic resources used in producing a product or providing a service. Different methods of costing are applied to different industries depending upon the type of manufacture and their nature. Broadly the costing methods are classified into the following:

- (a) Specific Order Costing (Job or Terminal Costing)

(b) Operation Costing or Process or Period Costing

Specific Order Costing: Specific order costing is the category of basic costing methods applicable where the work consists of separate jobs, batches or contracts each of which is authorised by a specific order or contract. It includes job costing consisting batch costing and contract costing.

Job Order Costing: CIMA London defines job costing as "the category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches each of which is authorised by specific order or contract".

Costing a job, batches of product, or contract, involves aggregation of all the cost elements necessary for the final product. For factory working process, adequate production control records must be maintained. Each job must be given an order number which must be cited or identified at all stages of production.

Where an order for a very large quantity of output is placed, it should be divided into separate batches so that job costing can be made without waiting for the completion of the whole work. The batches are then to be numbered along the line of the number given to the large quantity order. For example, where the order number is 30, the separate batches of the job could be 30.1, 30.2, 30.3, etc. for distinction purpose.

Overhead Cost Absorption Methods in Job Costing

In costing a job, the total of the direct material cost, direct wages, and direct expense of the job must be taken. Overhead is then applied as a percentage of direct material cost, percentage of wages, percentage of prime cost, rate per unit or rate per hour as required. The direct materials, wages and expenses, and the absorbed overhead are totalled to give the production cost of the job. Each job must absorb, in addition to production overhead, part of the administration, selling and distribution overheads, unless they are regarded as direct charges to the Profit and Loss Account of the period. The addition of selling, administrative and distribution overhead costs to production cost gives the total cost of the job.

In order to absorb overhead, it is necessary to develop a basis upon which overhead costs can be charged to cost unit. In general, the following arbitrary bases are used:

1. Percentage Rate on Direct Labour Cost, Direct material Cost and Prime Cost.
2. Cost unit Rate
3. Time Rate: Machine Hour Rate and Direct Labour Hour Rate.

ILLUSTRATION 1

Classic Modern Shoemakers Company Limited, Michika, found that their production costs for the 2000 accounting year had been as follows:

Direct wages N8,000,000

Direct material cost N5,000,000

Direct expenses N1,500,000

Production overhead cost N2,400,000

Total shoe manufactured 160,000 units Machine time used 10,000 hours Direct Labour time used 40,000 hours

500 pairs of shoes were produced during the first two weeks of the year as a single job, which had a direct wages cost of N8,000, direct material cost of N5,000 and direct expenses of N1,500. The company apportions fixed overhead cost using the percentage rate on direct wages and charges administrative expenses and selling expenses at the rate of 15% and 10% respectively, of production cost.

Required: Determine the following:

- (a) Production cost
- (b) Total cost
- (c) Cost per pairs of shoe

Illustration 2

Assume that the Shoe Manufacturing Company in illustration 1 apportions fixed overhead cost using the percentage rate on direct material cost, and all other figures remain the same, determine i – iii above.

Workings:

1. % Rate on Direct Material Cost = $\text{N}2,400,000 \times 100 = 80\% \text{N}3,000,000$
2. Production Overhead on the job = $80\% \text{ of N } 5,000 = \text{N}4,000$
3. Administrative expenses = $15\% \text{ of N}18,500 = \text{N}2,775$
4. Selling expenses = $10\% \text{ of } 18,500 = \text{N}1,850$

Computation of Production Cost & Total Cost

	#
Direct wages	8,000
Direct material cost	5,000
Direct expenses	1,500
Prime cost	14,500
Production overhead	4,000
Production Cost	18500 (i)
Administrative expenses	2,775
Selling expenses	1,850
Total Costs	<u>23,125</u> (ii)

- (iii) Cost per pair of Shoes = $\text{N}23,125 \div 500 = \text{N}46.25$

Batch Costing

Batch Costing arises where a customer orders a quantity of identical products at the same time. The job consists of a number of units of similar products covered by a single job number: for example, a batch of 1000 copies of a book printed by a printer. The batch cost must be divided by the total quantity to give a cost per unit.

Peculiar features of Batch Costing System:

- (a) *Set-Up Costs*: This is the cost of setting up production facilities, for example, the Engineering time, Factory consumables to set up moulds, etc.
- (b) *Unit Cost*: Average cost per unit will tend to decrease as the batch size increases due to some fixed costs (such as set up costs mentioned above) which will remain the same irrespective of the batch size.
- (c) *Batches of Dissimilar Products*: Sometimes batches of different products are brought together because a common operation can be performed on them so as to save set-up time on the machine.

Illustration 1

Ayngba Press Limited was asked to quote for supplying 5,000 and 25,000 booklets. The company normally expects a profit of 10% on sales. Costs were recognized as follows:

Paper and other materials, N (per 1,000 copies) 30.00
Wages (per 1,000 copies) 20.00
Layout cost 500.00
Fixed overhead 200.00
Variable overhead = 12% of wages

- (a) Draft a cost computation, showing minimum selling prices that might be quoted per 1,000 copies each of the supplies.

Solution

Calculation of Batch Cost

	1000 Copies #	5000 copies #	25000 copies #
Setup Cost or Lay-out Cost	500	500	500
Material Paper	30	150	750
Labour Wages	20	100	500
Overhead	200	200	200
	2.4	12	60
Fixed Overhead	752.40	962	2,010
Variable	83.60	107	223
Overhead Total	836	1069	2,233
Cost			
Profit Margin			
Selling Price			

ILLUSTRATION 2

XYZ manufacturers has the following budgeted overheads for the year based on normal activity levels.

Department	Budgeted overheads (#)	Overhead absorption base
Blanking	18000	1500 labour hours
Machining	43000	2500 machine hours
Welding	20000	1800 labour hours
Assembly	15000	1000 labour hours

Selling and administrative overheads are 20% of factory cost

An order for 250 assemblies type X 128, made as batch 5931, incurred the following cost:

Materials #3107

Labour 128 hours blanking shop at #5.25/hour

452 hours machining shop at #5.50/hour

90 hours assembly/shop at #4.80/hour

#525 was paid for the hire an equipment. The time booking in the machine shop was 643 machine hours.

You are required to calculate the total cost of the batch, the unit cost and the profit per assembly if the selling price was #150.

SOLUTION

The first step is to calculate the absorption rate for the production department

Blanking $\frac{\text{\#18000}}{\text{\#12 per labour hour}}$

1500

Machining #43000/2500 = #17.2 per machine hour
 Welding # 20000/1800 = # 11.1 per labour hour
 Assembly # 15000/1000 = #15 per labour hour

Total cost for batch No. 5931

	#	#
Direct material		3107.00
Direct expenses		525.00
Direct labour	128 x 5.25 452 x 5.50 90 x 5.25 175 x 4.80 =	<u>4470.50</u> 8102.50
Prime cost		

Production Overheads Absorption:

Blanking	128 x 12	
Machining	643 x 17.2	
Welding	90 x 11.1	
Assembly	175 x 15	=
Factory cost		<u>16219.60</u>
Add selling and admin exp:		24322.10
Overheads 20% of factory cost =		<u>4864.42</u>
= total cost		29186.52
Total cost/unit = 29186.52/250		
	= 116.75	

Therefore, profit/unit = #33.25 (selling price 150 – total cost/per unit 116.75)

CONTRACT COSTING

Contract Costing or Terminal Costing as it is often termed, is a variant of the job costing system, which is applied in businesses engaged in building or other construction work. The jobs are usually the contracts entered into with the customers. As the number of such contracts handled at a time by a business may not be usually large, Contract Costing is comparatively simpler in operation than job costing system. The basic principles applied in Contract Costing are the same as those used in job costing except that these are modified to suit the particular requirements of the contracts.

The person who takes contract for a price is called the **Contractor** and the person from whom it is taken is called the **Contractee**. We are mainly concerned with the books of the contractor. To find out profit earned or loss incurred on the contract, the contractor prepares a nominal account in his books called 'Contract Account'. In this account, all the expenses incurred by the contractor are debited and the income i.e. mainly work certified is credited; the difference represents profit or loss.

The contracts run for or number of years; however it is necessary to find out the profit or loss at the end of every year. The profit earned on a Contract Account is primarily called **Notional Profit** and a portion of which would be kept on reserve against contingencies

Differences between Job costing and Contract costing

- While the number of jobs in hand at any time in a concern may be large, only a few contracts may be undertaken at a time.
- The accumulation, analysis, apportionment, allocation and control of costs is simplified in Contract Costing.

(c) Most of the expenses are chargeable direct to the Contract Account. Direct allocation to such an extent is not possible in job costing.

(d) As contracts may run for long periods, there arises the problem of assessment and crediting of profits on incomplete contracts at the end of the accounting period. Contract can be classified into three categories:

- Contract at early stage
- Contract that have reached an advance stage but for which cost to complete the project cannot be reasonably estimated.
- Contract that are nearing completion that is, have reached an advance stage and for which cost is somehow complete and can be reasonably estimated.

Some special items under contract accounting:

(i) Sub-Contract

Sub-contracting may be necessary under the following circumstances:

- (a) Work of a specialized nature for which facilities are not internally available within the concern is offered to a sub-contractor.
 - (b) It may be advantageous to get a part or component from outside, if it is costlier to manufacture it.
 - (c) The capacity of the firm may be limited and in order to keep time schedule, work may be speeded by offering it to sub-contractors.
- The payments made to sub-contractors are charged in totals to the concerned Contract Account as direct expense and no detailed records or break-up of the sub-contract amount is necessary for cost purposes.

(ii) Surveyor's Certificate and Retention Money

In the case of contracts running for long periods of time, it is customary for the contractor's firm to get 'on account' payments against the portion of contract completed. The amount received depends upon the extent of work certified by the technical assessor i.e. on the surveyor's certificates, as these are called. Normally such payments are not received to the full extent of the work completed but a small percentage is held back as retention money, payable on completion of the contract. The retention money is a sort of safeguard available to the contractee in case the contractor is not able to fulfill one or more of the conditions laid down in the contract.

(iii) Defective Work

Defective work will not evidently be paid for by the contractee but the cost of such defective work should be charged to the Contract Account.

(iv) Escalation Clause

Escalation clauses are often provided in contracts as safeguards against any likely changes in price or utilisation of material and labour. Such a clause in a contract would provide that in the event of a specified contingency happening, the contract price would be suitably enhanced. This clause is particularly necessary where the price of certain raw materials are likely to rise, where labour rates are anticipated to increase.

(v) Work-in-progress

In Contract Accounts, the value of the work-in-progress consists of:-

- (a) the cost of work completed, both certified and uncertified,
- (b) the cost of work not yet complete, and
- (c) the amount of profit taken as credit.

(vi) Profit on incomplete contracts

For the purpose of finding out the portion of the profit out of notional profit to be transferred to Profit and Loss Account, the contracts are divided in the following manner:-

- Contracts which have just commenced: the contracts in which less than 1/4th work is done are regarded as the contracts which have just commenced.
- Contracts which have reasonably advanced: the degree of completion of work is $> 1/4$ and $< 1/2$.

- Contracts which are almost complete

Progress payments

The contract normally provides for the client to make progress payments either at specific stages of the work eg when foundations are completed, or at particular agreed interval. The basis for this interim payment is an architect's certificate of work satisfactorily completed.

Illustration 1

An architect assesses the value of work done to be #265000. The client has already paid #110,000 and the agreed retention percentage is 15%. What is the amount of the current progress payment?

Solution

$$\begin{aligned}\text{Current payment} &= \text{value certified} - \text{Retention} - \text{Payments already made} \\ &= \#265,000 - (15\% \text{ of } 265000) - 110,000 \\ &= 265,000 - 39,750 - 110,000 \\ &= \#115,250\end{aligned}$$

The accounting entries for this payment when received will be

Dr Bank account

Cr Client account for cash received on account.

When estimating interim profit :

a. if a substantial cost have been incurred eg 30% - 80% completed the formula for calculating profit is
= 2/3 or 3/4 of the notional profit x cash received from progress payment

Value of work certified

Where the notional profit is Value of work Certified – cost of work Certified.

b. When the contract is nearing completed eg above 80%, one of the following methods may be used :

Profit taken = progress payments to date x estimated total pft on completion

Contract price

OR Profit taken = Value of work certified – cost of work certified

Illustration 2

At the end of year 2022 Apex Developments has three contracts in progress and their details are as follows:

CONTRACT	AP10	AP11	AP12
	#	#	#
Contract price	150,000	275,000	185,000
Costs to date	35,000	144,000	154,000
Estimated cost of completion	88,000	96,000	7000
Value of work certified	40,000	165,000	172,000
Progress payments received	34,000	140,250	146,200
Cost of work certified	28,000	138,000	150,000

Interim profits if any, should be taken on the three contracts. (no profits have been taken so far)

Solution

First check the degree of completion and whether the contracts are expected to make a completion

	AP10	AP11	AP12
#	#	#	#
Contract price	150,000	275,000	185,000
Less: estimated total cost	<u>123,000</u>	<u>240,000</u>	<u>161,000</u>
Estimated contract profit	<u>27,000</u>	<u>35,000</u>	<u>24,000</u>

Approximate degree of completion:

<u>Cost to date</u>			
Total costs	=	<u>35,000</u>	<u>144,000</u>
		123,000	240,000
		= 28%	60%

Thus all contract are expected to make an overall profit on completion so taking an interim profit can be considered provided that the degree of completion justifies doing only 28% complete so it would be prudent not to take any profit until more of the contract is completed:
Contract AP11

$$\begin{aligned} \text{Profit taken} &= \frac{2}{3} \times \text{Notional Profit} \times \frac{\text{Progress payments}}{\text{Value of work certified}} \\ &= \frac{2}{3} \times 165,000 - 138,000 \times \frac{140,250}{165,000} \\ &= \#15,300 \end{aligned}$$

$$\begin{aligned} \text{Contract AP 12} \\ \text{Profit taken} &= \frac{\text{Progress payments}}{\text{Contract price}} \times \text{Estimated total profit} \\ &= \frac{146,200}{185,000} \times 24,000 = \#18,966 \end{aligned}$$

PROCESS COSTING

Process Costing is a method used in a situation where production follows a series of sequential processes. The method is used to ascertain the cost of a product or service at each stage of production, manufacture or process.

Process costing is used in a variety of industries, including food processing, paper milling, chemical and drug manufacturing, oil refining, soap making, textiles, box-making, paint and ink manufacturing, brewery, flour milling, bottling and canning, biscuits products, meat products, sugar making, etc. It is probably the most widely used cost accounting system everywhere in the world.

Although, details will vary from one business concern to another, there are common features in most process costing systems that should be taken note of. These are:

- (a) Expenditure for each cost centre is collected and, at the end of the accounting period, the cost of the completed units are then transferred into a stock account or to a further process cost centre.
- (b) Accurate records are required of units produced and part-produced as well as the cost incurred by the cost centres.
- (c) The cost unit chosen should be relevant to the organisation.
- (d) The cost of the output of one process is the raw material input cost of the next process stage.
- (e) Wastage due to scrap, chemical reaction or evaporation is unavoidable. The operation or manufacturing should, however, be in such a way that wastage can be reduced to the barest minimum.

Process Losses and Wastages

In many industries, the amount of the process output will be less than the amount of the materials input. Such shortages are known as process losses or wastages, which may arise due to a variety of factors such as evaporation, scrap, shrinkage, unavoidable handling, breakages, etc.

If the losses are in accordance with normal practice they are known as normal process losses. But where losses are above expectation, they are known as abnormal losses, and as such they should be charged to an appropriate account pending investigation.

Normal process wastages are unavoidable losses arising from the nature of the production process and, so, it is logical and equitable that the cost of such losses is included as part of the cost of good production.

Illustration 1:

TMR produces 100 units from the input of 120 units from material that cost is N27, 000 and conversion cost of N13, 000. Normal process wastage was determined in advance as 5% of input. The output is to be sold at N150 per kg. Wastages is valued as N50=00 per kg.
Required to determine:

- c. Cost per unit of the expected output,
- d. Production cost of the output achieved,
- e. Cost of abnormal loss or abnormal gain obtained.

Solution 1

Units produced Cost

Input : Material	(kg)	#
Conversion	120	27,000
		<u>13,000</u>
		40,000
Less Normal wastage (5%)	(6)	<u>(300)</u>

Expected output,	114	39,700
Output	(100)	
Abnormal Loss	14	

a. Cost per kg of Expected output = $\frac{N39,700}{114} = N348.246$

b. Production cost of the output achieved: 100 units x N348.246 = N34,824.6

c. Cost of abnormal loss: 14 units x N348.246 = N4875.44
46 = N4875.44

Illustration 2

Using the information given in illustration 1, prepare Process account And abnormal loss account.

Solution 2

		<u>Process Account</u>			
	Kg	Value #		Kg	Value #
Material	120	27000	Good production	100	34,825
Conversion		13000	Normal wastage	6	300
	<u>120</u>	<u>40000</u>	Abnormal wastage	<u>114</u>	<u>4875</u>
				120	40000

Abnormal wastage Account

#	#
Process A/c 4,875	Profit & Loss 4875

EQUIVALENT UNITS: At the end of a given period, in the course of the production process, it is virtually certain that some items will only be partly completed (work in progress). Some of the costs of the periods are therefore attributable to these partly completed units as well as to those that are fully completed. In calculating equivalent units, it is more desirable to consider the percentage completion of each of the cost elements: material, labour and overhead. Here each cost element must be treated separately and then the costs per unit of each element are added to give the cost of a complete unit.

Illustration 3:

The production and cost data of Elsemco Shoemakers for the month of January, 2021 were as follows:

	#
Materials	422,400
Labour	395,600
Overhead	225,000
Total cost	1,043,000

Production was 8,000 fully completed units and 2,000 partly completed. The percentage completion of the 2,000 units' work-in-process was:

Material 80%

Labour 60%

Overhead 50%

Required: Find the value of completed production and the value of work-in-progress.

Solution 3

Cost	Equiv. Units	Fully compl.	Total prod.	Total cost/unit
Elements	in WIP	Units	Cost	
	#		#	
Material	80 % of 2000 = 1600	8000	9600	422,400 - 44
Labour	60% of 2000 = 1200	8000	9200	395,600 43
Overhead	50% of 2000 = 1000	8000	<u>9000</u>	<u>225,000</u> 25
			1,043,000	112

Value of completed units = # 112 x 8000 = #896,000

Value of WIP = TC - Value of completed units

= 1,043,000 - 896,000 = # 147,000

To check the value of WIP, the cost per each cost element is to be multiplied by the number of equivalent units of production in WIP related to each cost element.

Element	No. of equiv. units in WIP	Cost/unit	Value of WIP
Material	# 1,600	44	# 70,400
Labour	1,200	43	51,600
Overhead	1,000	25	<u>25,000</u>
			147,000

Element	Process Account		
	Unit cost	Total	Element
	#	#	Unit cost
			Total
			#

25

Material	10000	422,400		
Labour		395600	Goods transferred:	
Overhead		<u>225000</u>	to next stage	8000
	10000	<u>1,043,000</u>	WIP c/d	<u>2000</u>
WIP b/d	2000	147,000		10000
				<u>1,043,000</u>

Valuation Process and Cost of Production Report

A number of stages are passed through in the valuation necessary in process costing for cost statement.

- First, the physical flow of the units of production must be calculated having regards to the total number of units to be accounted for, regardless of the degree of completion.
 - Secondly, the equivalent units involved in the physical flow are to be calculated.
 - Thirdly, the total equivalent units and the current equivalent units involved are to be calculated. These are to be accounted for in respect of the cost elements (transferred-in cost, material cost and conversion cost).
 - Fourthly, the unit costs are to be calculated, paying regards to the stock valuation method assumed (FIFO, WAP, LIFO, etc.).
 - Finally, the total cost of the transferred-out products and work-in-process are to be calculated, ensuring that all costs are accounted for.
- The cost of production report is to show the number of units of output to be accounted for, the total equivalent units of completed output, the cost statement showing the impact of all the cost elements and the cost of completed units as well as that of the work-in-process at the end of the reporting period.